

SVHN Classification – Configuration Comparison Report

Configuration A — Baseline

Settings: No data augmentation, SGD optimizer, learning rate 0.01, weight decay $1e-3$, label smoothing 0.1 and 20 epochs .

Results:

Train Accuracy: **99.95%**

Test Accuracy: **60.4%**

Runtime: **7.22 minutes**

Analysis:

The model quickly overfit the training data due to lack of augmentation or regularization. While training accuracy was near perfect, test accuracy remained low, showing poor generalization.

Configuration B — Light Augmentation + Adam

Settings: Light augmentations (flip + crop), Adam optimizer, learning rate 0.001, weight decay $1e-4$, label smoothing 0.1 and 30 epochs .

Results:

Train Accuracy: **75.3%**

Test Accuracy: **62.5%**

Runtime: **18.16 minutes**

Analysis:

Light augmentations improved generalization, and the Adam optimizer provided smoother convergence. However, the model underfit slightly (low training accuracy), indicating that learning rate or weight decay could be tuned further.

Configuration C — Stronger Augmentation + SGD

Settings: Stronger color and rotation augmentations, SGD optimizer, learning rate 0.02, weight decay $5e-4$, label smoothing 0.1 and 25 epochs.

Results:

Train Accuracy: **99.67%**

Test Accuracy: **67.66%**

Runtime: **26.42 minutes**

Analysis:

Stronger augmentations helped prevent overfitting, resulting in a +5% test accuracy gain compared to Config B. Using SGD with momentum improved stability and generalization, while moderate label smoothing enhanced robustness.

Configuration D — Test-Time Augmentation (TTA)

Settings: Same augmentations as C, TTA (flip + rotation), higher label smoothing 0.3, weight decay 1e-3, learning rate 0.015 and 30 epochs.

Results:

Train Accuracy: **99.80%**

Test Accuracy: **70.0%**

Runtime: **33.13 minutes**

Analysis:

Test-time augmentation boosted performance by averaging predictions over multiple image transformations, improving robustness to viewpoint changes. Despite slightly longer runtime, this setup achieved the best generalization among all configurations.